Functional Analysis-I

Tsinghua University, 2019-11-08

Name:_____ Class: _____ Student ID: _____

1. (10%) Let $C[0,1] := \{f : [0,1] \to \mathbb{R} : f \text{ is continuous}\}$ with norm $\|f\| = \max_{x\in[0,1]} |f(x)|$. Define $T : C[0,1] \to C[0,1]$ by

$$T(f)(t) = \int_0^t sf(s)ds, \quad t \in [0,1].$$

Show that for any $n \in \mathbb{N}$, $\|T^n\| = \frac{1}{2^n n!}$, where $T^n = T \circ T \circ \cdots \circ T$.

Proof. Clearly $|T(f)(t)| \le \frac{1}{2}\|f\|t^2$, so

$$|T^2(f)(t)| = \left|\int_0^t sT(f)(s)ds\right| \le \int_0^t \frac{1}{2}\|f\|s^3 ds = \frac{1}{4\times 2}\|f\|t^4.$$

By induction we have $|T^n(f)(t)| \le \frac{1}{2^n n!}\|f\|t^{2n}$. This implies $\|T^n\| \le \frac{1}{2^n n!}$. Conversely let $f(t) \equiv 1$, we easily obtain $T^n(f)(t) = \frac{1}{2^n n!}t^{2n}$ and so

$$\|T^n\| \ge \|\tfrac{1}{2^n n!}t^{2n}\| = \tfrac{1}{2^n n!}.$$

□

2. (10%) Let X, Y be two normed vector spaces. Given $x_0 \in X \setminus \{0\}$ and $y_0 \in Y$. Show that there is $A \in \mathcal{L}(X,Y)$ such that $A(x_0) = y_0$.

Proof. Define $f : \mathbb{R}x_0 \to \mathbb{R}$ by

$$f(ax_0) = a.$$

Then f is linear and continuous. By the Hahm-Banach Theorem, there is $F \in X^*$ such that $F(ax_0) = f(ax_0) = a$. Letting $A(x) = F(x)y_0$ we conclude $A(x_0) = y_0$ and A is linear continuous. □

3. (10%) Let E be an infinite dimensional normal vector space, and let $f : E \to \mathbb{R}$ be linear but not continuous.

(1) (5%) Show that $f^{-1}(1) := \{x \in E : f(x) = 1\}$ is dense in E.

(2) (5%) Construct 2 convex subsets C_1, C_2 of E such that $C_1 \cap C_2 = \emptyset$, $C_1 \cup C_2 = E$, and both C_1 and C_2 are dense in E.

Proof. (1) We show that for any $\lambda \in \mathbb{R}$, $f^{-1}(\lambda) := \{x \in E : f(x) = \lambda\}$ is dense in E. Since f is linear and not continuous, so f is not continuous at $x = 0$, i.e. there is $(x_n)_{n\geq 1} \subset E$ and constant $c_0 > 0$ such that

$$\lim_{n\to+\infty} x_n = 0 \quad \text{but} \quad |f(x_n)| \geq c_0 > 0.$$

By replacing x_n by $x_n/f(x_n)$, we can assume

$$\lim_{n\to+\infty} x_n = 0, \quad f(x_n) = 1.$$

For any $x \in E$, if $f(x) = \lambda$, then $x \in f^{-1}(\lambda)$. If $f(x) \neq \lambda$, then letting $y_n = x + (\lambda - f(x))x_n$, we have $f(y_n) = f(x) + (\lambda - f(x))f(x_n) = \lambda$, i.e. $y_n \in f^{-1}(\lambda)$ and $y_n \to x$. This proves that $f^{-1}(\lambda) := \{x \in E : f(x) = \lambda\}$ is dense in E.

(2) Letting $C_1 = \{x \in E : f(x) \leq 0\}$ and $C_2 = \{x \in E : f(x) > 0\}$, we see from the linearity of f that C_1, C_2 are convex, $C_1 \cap C_2 = \emptyset$ and $C_1 \cup C_2 = E$. Since $f^{-1}(0) \subset C_1$ and $f^{-1}(1) \subset C_2$, both C_1 and C_2 are dense in E. □

4. (10%) Let X, Y, Z be Banach spaces. Let $T \in \mathcal{L}(X, Z)$ and $A \in \mathcal{L}(Y, Z)$. Suppose for any $x \in X$, there is a unique $y \in Y$ such that $A(y) = T(x)$. Define $B : X \to Y$ by $B(x) = y$. Show that B is linear and continuous, namely $B \in \mathcal{L}(X, Y)$.

Proof. First show that B is linear. For any x_1, x_2, denote $y_i = B(x_i)$. Then $A(y_i) = T(x_i)$. By the linearity of A, T, for any $\alpha, \beta \in \mathbb{R}$,

$$T(\alpha x_1 + \beta x_2) = \alpha T(x_1) + \beta T(x_2) = \alpha A(y_1) + \beta A(y_2) = A(\alpha y_1 + \beta y_2),$$

so by the definition of B, we have

$$B(\alpha x_1 + \beta x_2) = \alpha y_1 + \beta y_2 = \alpha B(x_1) + \beta B(x_2).$$

Second, show that B is closed. Suppose $(x_n, B(x_n)) \to (x, y)$ in $X \times Y$, we need to prove $y = B(x)$. Clearly $T(x_n) = A(y_n)$, where $y_n := B(x_n) \to y$. Then by the continuity of T, A we have

$$T(x) = \lim_{n\to+\infty} T(x_n) = \lim_{n\to+\infty} A(y_n) = A(y).$$

So the definition of B implies $y = B(x)$. This shows that B is closed, so the Closed Graph Theorem implies that $B \in \mathcal{L}(X, Y)$. □

5. (12%) Let

$$E = \left\{ x = \{\xi_k\}_{k\geq 1} : \xi_k \in \mathbb{R}, \lim_{k\to\infty} \xi_k \text{ exists} \right\} \quad \text{with} \quad \|x\| := \sup_k |\xi_k|.$$

(1) (6%) Show that E is a Banach space.

(2) (6%) Given a sequence $\{\alpha_k\}$ of real numbers. If $\sum_{k=1}^{\infty} \alpha_k \xi_k$ is convergent for each $x = (\xi_k)_{k\geq 1} \in E$, show that

$$\sum_{k=1}^{\infty} |\alpha_k| < \infty.$$

Proof. (1) Let $x_n = (\xi_{n,k})_{k\geq 1}$, $n \in \mathbb{N}$ be a Cauchy sequence in E, i.e.

$$\|x_n - x_m\| \to 0 \quad \text{as} \quad n, m \to +\infty.$$

Since for any $k \geq 1$,

$$|\xi_{n,k} - \xi_{m,k}| \leq \|x_n - x_m\|,$$

so $(\xi_{n,k})$, $n \geq 1$ is a Cauchy sequence in $\mathbb{R}$. So $\xi_k := \lim_{n\to\infty} \xi_{n,k}$ is well-defined. Define

$$x = (\xi_k)_{k\geq 1},$$

we need to prove $x \in E$ and $\|x_n - x\| \to 0$ as $n \to +\infty$.

Given any $\varepsilon > 0$, there is $N = N_\varepsilon \in \mathbb{N}$ such that

$$|\xi_{n,k} - \xi_{m,k}| \leq \|x_n - x_m\| \leq \varepsilon, \quad \forall n, m \geq N, \quad \forall k \geq 1.$$

Letting $m \to +\infty$, we obtain

$$|\xi_{n,k} - \xi_k| \leq \varepsilon, \quad \forall n \geq N, \quad \forall k \geq 1.$$

First this implies $\|x_n - x\| = \sup_k |\xi_{n,k} - \xi_k| \leq \varepsilon$ for any $n \geq N$, so $\|x_n - x\| \to 0$ as $n \to +\infty$.

Second, since $x_N \in E$ implies $\lim_{k\to+\infty} \xi_{N,k}$ exists, so there exists $M \in \mathbb{N}$ such that

$$|\xi_{N,k} - \xi_{N,l}| \leq \varepsilon, \quad \forall k, l \geq M.$$

Therefore, for any $k, l \geq M$,

$$|\xi_k - \xi_l| \leq |\xi_k - \xi_{N,k}| + |\xi_{N,k} - \xi_{N,l}| + |\xi_{N,l} - \xi_l| \leq 3\varepsilon,$$

namely (ξ_k), $k \geq 1$ is a Cauchy sequence in $\mathbb{R}$, so $\lim_{k\to+\infty} \xi_k$ exists. This implies $x \in E$.

(2) Define $f_n : E \to \mathbb{R}$ by

$$f_n(x) = \sum_{k=1}^{n} \alpha_k \xi_k, \quad x = (\xi_k)_{k\geq 1} \in E.$$

It is easy to see that $f_n \in E^*$ and

$$\|f_n\| = \sum_{k=1}^{n} |\alpha_k|.$$

Since for any $x \in E$, $f_n(x) \to \sum_{k=1}^{\infty} \alpha_k \xi_k$, we can apply the Uniform Boundedness Theorem to conclude that

$$\sup_n \|f_n\| < +\infty,$$

so $\sum_{k=1}^{\infty} |\alpha_k| < \infty$. □

6. (12%) (1) (4%) Write down the statements of the closed graph theorem and the inverse operator theorem, and

(1) (4%) Prove the closed graph theorem by using the inverse operator theorem;

(2) (4%) Prove the inverse operator theorem by using the closed graph theorem.

Proof. Closed graph theorem: Let E, F be two Banach spaces. Let T be a linear operator from E into F. Assume that the graph of T, $G(T)$, is closed in $E \times F$, then T is continuous.

Inverse operator theorem: Let E, F be two Banach spaces and let T be a continuous linear operator from E into F that is bijective. Then T^{-1} is also continuous.

(1) Let $T : E \to F$ be a closed linear operator, namely $G(T)$ is a closed subspace of $(E \times F, \|\|_E + \|\|_F)$, so $G(T)$ is a Banach space. Define $A : G(T) \to E$ by

$$A([x, Tx]) = x.$$

Then A is linear continuous and A is bijective. So the inverse operator theorem implies that A^{-1} is also continuous, so

$$\|Tx\| \leq C\|x\|.$$

(2) Let $T : E \to F$ be a bounded linear bijective operator, and $T^{-1} : F \to E$ be the inverse operator. It suffices to show that T^{-1} is closed. Let

$$y_n \to y \quad T^{-1}(y_n) \to x,$$

we need to show $x = T^{-1}(y)$. In fact, since T is continuous, we have $y_n = T(T^{-1}(y_n)) \to T(x)$ and so $y = T(x)$, which gives $x = T^{-1}(y)$. By the Closed graph theorem, we obtain that $T^{-1} \in \mathcal{L}(F, E)$. □

7.(12%) Let A be a convex subset of a Banach space E. Suppose that A is separable for the weak topology $\sigma(E, E^*)$. Show that A is separable for the norm topology.

Proof. Let $\{x_n\}_{n \geq 1} \subset A$ be a dense subset of A for the weak topology. Define

$$B := \left\{ \sum_n \lambda_n x_n : \lambda_n \geq 0, \sum_n \lambda_n = 1, \lambda_n \neq 0 \text{ for finite } n\text{'s} \right\}$$

to be the convex set generated by $\{x_n\}_{n\geq 1}$. Since A is convex, so $B \subset A$. Since

$$B_{\mathbb{Q}} := \left\{ \sum_n \lambda_n x_n : 0 \leq \lambda_n \in \mathbb{Q}, \sum_n \lambda_n = 1, \lambda_n \neq 0 \text{ for finite } n\text{'s} \right\}$$

is countable, and a norm dense subset of B, it suffices to prove $A \subset \overline{B}^{\|\|}$.

Indeed, for any $x \in A$, since $\{x_n\}_{n\geq 1} \subset A$ is dense in A for the weak topology, there is a subsequence $x_{n_k} \rightharpoonup x$ weakly. Then by Mazur's corollary, there is a sequence (y_n) made up of convex combinations of x_{n_k}'s, i.e. $y_n \in B$ such that $y_n \to x$ strongly. This proves $A \subset \overline{B}^{\|\|}$. □

8. (12%) Let E be a reflexive Banach space, and let $\{f_0, f_1, \cdots, f_n\} \subset E^*$. Suppose

$$[f_i(x) \geq 0,\ \forall i = 1, \cdots, n] \Longrightarrow f_0(x) \geq 0.$$

Show that there are $\lambda_1, \cdots, \lambda_n \geq 0$ such that $f_0 = \sum_{j=1}^n \lambda_j f_j$.

Proof. Let

$$C = \left\{ \sum_{j=1}^n \lambda_j f_j \ :\ \lambda_j \geq 0, \forall j = 1, \cdots, n \right\}.$$

Then C is a closed, convex subset in E^*. Suppose $f_0 \notin C$, then by the second geometric form of the Hahm-Banach Theorem, there is $\varphi \in E^{**}$ and $\alpha \in \mathbb{R}$ such that

$$\langle \varphi, f_0 \rangle < \alpha < \langle \varphi, f \rangle, \quad \forall f \in C.$$

Since E is reflexive, there is $x \in E$ such that $\varphi = Jx$, so

$$f_0(x) < \alpha < f(x), \quad \forall f \in C.$$

Clearly $0 \in C$ implies $f_0(x) < \alpha < 0$. For any $\lambda > 0$, we have $\lambda f_j \in C$ and so $f_j(x) > \frac{\alpha}{\lambda}$. Letting $\lambda \to +\infty$ we obtain $f_j(x) \geq 0$ for all $j = 1, \cdots, n$, so $f_0(x) \geq 0$, a contradiction. This proves $f_0 \in C$. □

9.(12%) Let E, F be Banach spaces, with F reflexive and E separable. If $\{A_k\}_{k=1}^{\infty} \subset \mathcal{L}(E, F)$ satisfies

$$\sup_{k\geq 1} \|A_k\| \leq M \quad \text{for some } M \in \mathbb{R}.$$

Show that there is a subsequence $A_{k,k}$ such that $A_{k,k}x$ converges weakly in $\sigma(F, F^*)$ for each $x \in E$ as $k \to \infty$.

Proof. Let $\{x_n\}_{n\geq 1}$ be a dense subset of E. Since $A_k x_1$ is a bounded sequence in F and F is reflexive, so there is a subsequence

$$\{A_{1,1}, A_{1,2}, \cdots, A_{1,k}, \cdots\} \subset \{A_1, A_2, A_3 \cdots\}$$

such that $A_{1,k}x_1 \rightharpoonup y_1$ for some $y_1 \in F$. Since $A_{1,k}x_2$ is a bounded sequence in F, so there is a subsequence

$$\{A_{2,1}, A_{2,2}, \cdots, A_{2,k}, \cdots\} \subset \{A_{1,1}, A_{1,2}, \cdots, A_{1,k}, \cdots\}$$

such that $A_{2,k}x_2 \rightharpoonup y_2$ for some $y_2 \in F$.

By induction, for any $n \in \mathbb{N}$, there is a subsequence

$$\{A_{n,1}, A_{n,2}, \cdots, A_{n,k}, \cdots\} \subset \{A_{n-1,1}, A_{n-1,2}, \cdots, A_{n-1,k}, \cdots\}$$

such that $A_{n,k}x_n \rightharpoonup y_n$ as $k \to \infty$ for some $y_n \in F$.

Consider the subsequence $\{A_{k,k}\}_{k=1}^{\infty}$. Since for any n, we have

$$\{A_{n,n}, A_{n+1,n+1}, \cdots\} \subset \{A_{n,1}, A_{n,2}, \cdots, A_{n,k}, \cdots\},$$

we have $A_{k,k}x_n \rightharpoonup y_n$ as $k \to \infty$.

Now for any $x \in E$, we want to prove $A_{k,k}x$ weakly converges to some y, i.e. to prove that for any $f \in F^* \setminus \{0\}$,

$$f(A_{k,k}x) \to f(y) \quad \text{as } k \to \infty.$$

Since $\{x_n\}_{n\geq 1}$ is dense in E, there is a subsequence, still denoted by $\{x_n\}_{n\geq 1}$ just for convenience, such that

$$x_n \to x.$$

Since for any n, m, $A_{k,k}(x_n - x_m) = A_{k,k}x_n - A_{k,k}x_m \rightharpoonup y_n - y_m$ as $k \to \infty$, so

$$\|y_n - y_m\| \leq \liminf_{k\to\infty} \|A_{k,k}(x_n - x_m)\| \leq M\|x_n - x_m\|,$$

namely $\{y_n\}_{n\geq 1}$ is a Cauchy sequence, so $y_n \to y$ in F for some y.

For any $\varepsilon > 0$, there is $n_0 \in \mathbb{N}$ such that

$$\|x - x_{n_0}\| < \frac{\varepsilon}{3\|f\|M}, \quad \|y - y_{n_0}\| < \frac{\varepsilon}{3\|f\|}.$$

Since $A_{k,k}x_{n_0} \rightharpoonup y_{n_0}$ as $k \to \infty$, we have $f(A_{k,k}x_{n_0}) \to f(y_{n_0})$, so there exists $N \in \mathbb{N}$ such that

$$|f(A_{k,k}x_{n_0}) - f(y)| < \frac{\varepsilon}{3}, \quad \forall k \geq N.$$

Consequently, for any $k \geq N$,

$$\begin{aligned}|f(A_{k,k}x) - f(y)| &\leq |f(A_{k,k}(x - x_{n_0}))| + |f(A_{k,k}x_{n_0}) - f(y_{n_0})| + |f(y_{n_0}) - f(y)| \\ &\leq M\|f\|\|x - x_{n_0}\| + \frac{\varepsilon}{3} + M\|y_{n_0} - y\| \leq \varepsilon.\end{aligned}$$

This proves that $f(A_{k,k}x) \to f(y)$ and so $A_{k,k}x \rightharpoonup y$. □

10. (10%) Let $\{u_n\}$ be a sequence in a Hilbert space H which converges weakly to $u \in H$, namely

$$(u_n, v) \to (u, v) \quad \forall v \in \mathbb{H}.$$

Show that there is a subsequence $\{v_k\}$ of $\{u_n\}$ such that $(v_1 + ... + v_k)/k$ converges strongly to u in H as $k \to +\infty$.

Proof. Since $u_n \rightharpoonup u$, so there exists $C > 0$ such that

$$\|u_n\|^2, \|u\|^2, |(u_n, u)| \le C, \quad \forall\, n.$$

Let $v_1 = u_1$. Since $(v_1, u_n) \to (v_1, u)$, so there exists $N_2 \ge 2$ such that

$$|(v_1, u_n) - (v_1, u)| < 1/2, \quad \forall\, n \ge N_2.$$

Let $v_2 = u_{N_2}$. Since $(v_2, u_n) \to (v_2, u)$, so there exists $N_3 \ge N_2 + 1$ such that

$$|(v_2, u_n) - (v_2, u)| < 1/2^2, \quad \forall\, n \ge N_3.$$

Let $v_3 = u_{N_3}$. By an induction procedure, we obtain a subsequence $\{v_n : n \ge 1\}$ of $\{u_n\}$ such that

$$|(v_i, v_j) - (v_i, u)| < \frac{1}{2^i}, \quad \forall\, j > i. \tag{0.1}$$

We should prove that $\frac{1}{k}\sum_{i=1}^k v_i \to u$. Recall that $v_n \rightharpoonup u$, so for any $\varepsilon > 0$, there exists N such that

$$|(v_n, u) - \|u\|^2| < \varepsilon, \quad \forall\, n \ge N.$$

For any $k > N$ we have

$$\begin{aligned}
&\left\| \frac{1}{k}\sum_{i=1}^k v_i - u \right\|^2 = \frac{1}{k^2}\left| \sum_{i,j=1}^k (v_i, v_j) - 2k\sum_{i=1}^k (v_i, u) + k^2\|u\|^2 \right| \\
\le &\frac{1}{k^2}\left| \sum_{i=1}^k \|v_i\|^2 + 2\sum_{i<j}^k (v_i, v_j) - 2k\sum_{i=1}^k (v_i, u) + k^2\|u\|^2 \right| \\
\le &\frac{C}{k} + \frac{1}{k^2}\left| 2\sum_{i=1}^{k-1}\sum_{j=i+1}^k [(v_i, v_j) - (v_i, u)] - 2\sum_{i=1}^k i(v_i, u) + k^2\|u\|^2 \right| \\
\le &\frac{C}{k} + \frac{1}{k^2} 2\sum_{i=1}^{k-1}\sum_{j=i+1}^k \frac{1}{2^i} + \frac{1}{k^2}\left| k^2\|u\|^2 - 2\sum_{i=N+1}^k i(v_i, u) - 2\sum_{i=1}^N i(v_i, u) \right| \\
\le &\frac{C+2}{k} + \frac{2}{k^2}\sum_{i=1}^N i|(v_i, u)| \\
&+ \frac{1}{k^2}\left| 2\sum_{i=N+1}^k i[\|u\|^2 - (v_i, u)] + [k^2 - (k-N)(k+N+1)]\|u\|^2 \right| \\
\le &\frac{C+2}{k} + \frac{N(N+1)C}{k^2} + \frac{2}{k^2}\sum_{i=N+1}^k i\varepsilon + \frac{k + N(N+1)}{k^2} C \\
= &\frac{C+2}{k} + \frac{N(N+1)C}{k^2} + \frac{(k-N)(k+N+1)}{k^2}\varepsilon + \frac{k + N(N+1)}{k^2} C.
\end{aligned}$$

Hence,

$$\limsup_{k\to\infty} \left\| \frac{1}{k} \sum_{i=1}^{k} v_i - u \right\|^2 \leq \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, so

$$\limsup_{k\to\infty} \left\| \frac{1}{k} \sum_{i=1}^{k} v_i - u \right\|^2 = 0.$$

This means $\frac{1}{k} \sum_{i=1}^{k} v_i \to u$. □